The Timing of Early-Life Shocks and Adolescent Mental Health

October 2026

Do the effects of an earthquake in early childhood persist?



Shock 27 February 2010 (27-F), magnitude 8.8

Exposed Children aged 6 months to 4 years

Effects of early-childhood shocks

Birth weight (Torche, 2011)

Cognitive and non-cognitive development (Gillmore, 2026)

Height and learning (Andrabi, Daniels, and Das, 2023)

Adult mental health (Persson and Rossin-Slater, 2018)

This paper

Mental health at ages 14–18 by age at the earthquake, from 6 to 49 months

The 2024 round follows the 2010 cohort, born by August 2009: exposure in utero or before 6 months is not observed

▸ PGA map

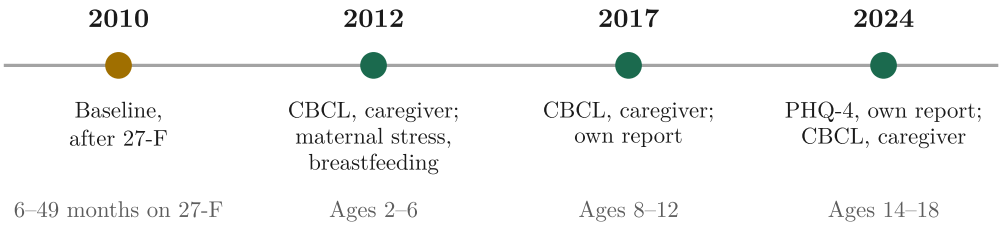
Does it matter at what age children live through the shock?

> "The effects of stress on the brain depend critically on the timing (age of onset and duration). When stress occurs early in life it can have profound and lasting effects on brain organization and function." (Gee and Casey, 2015, p. 184)

Potential mechanisms

	Channels
Socioeconomic and family environment	The earthquake reduces income, damages housing, and increases conflict at home (Conger and Donnellan, 2007; Almond, Currie, and Duque, 2018)
Transmission of maternal stress	Infants regulate their stress through the mother. If she is stressed, she passes it on through daily contact (Persson and Rossin-Slater, 2018)

Data



Survey	ELPI, Early Childhood Longitudinal Survey, Chile
Sample	9,996 adolescents born between January 2006 and August 2009, aged 6 to 49 months on 27-F; 4,961 girls
Respondents	The caregiver in every round; the child in 2017 and 2024

▸ Descriptives

Econometric specification

$$Y_{imc} = \sum_a \beta_a \left(\text{Age}_{ai} \times \text{Affected}_m\right) + \sum_a \gamma_a \,\text{Age}_{ai} + \psi_m + \theta_c + X_i'\delta + \varepsilon_{imc}$$

Y_{imc}	Mental health at ages 14–18, PHQ-4
Age_{ai}	Six-month age groups at the earthquake: 6–11, 12–17, …, 42–49 months
Affected_m	Municipality in the zone of destructive damage
ψ_m, θ_c	Municipality and birth-year fixed effects
X_i	Linear trend in month of birth; child and household controls

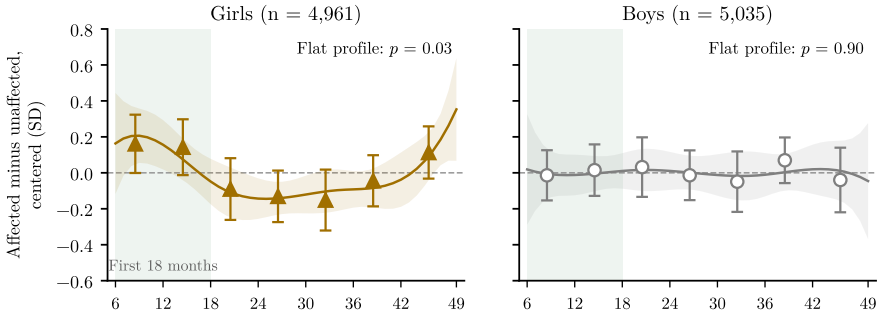
Profile β_a normalized to average zero; joint test of a flat profile

Summary First 18 months relative to all other ages, chosen after inspecting the profile

Affected_m does not vary within municipality, so ψ_m absorbs the average effect of the earthquake: the design identifies how the effect varies with age at exposure, not its level. Replacing γ_a, θ_c, and the trend with month-of-birth fixed effects leaves the estimates unchanged.

▸ Balance ▸ Attrition

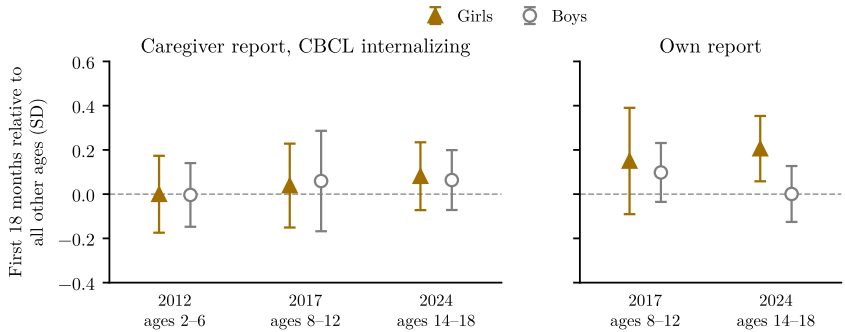
Age at exposure matters for girls, not for boys



PHQ-4 (Kroenke et al., 2009), four questions answered by the adolescent on anxiety and depression. Affected minus unaffected in each six-month age group, normalized to average zero; the curve is a B-spline in age in months; 95% CI. Summary comparison, chosen after inspecting the profile: the first 18 months relative to all other ages, 0.21 SD for girls (SE 0.08; wild bootstrap $p = 0.01$; randomization across regions $p = 0.06$) and 0.00 for boys; girls minus boys, 0.21 (SE 0.10).

▸ Distribution ▸ Levels ▸ Table ▸ Other outcomes

Girls report the gap; their caregivers do not

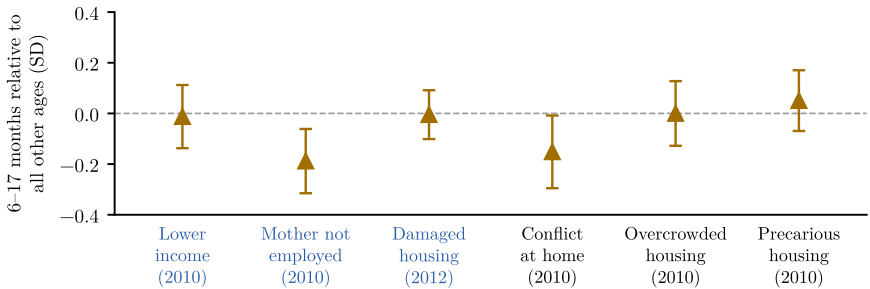


The same adolescents in each round, with the specification of the previous slide; 95% CI. Caregiver report, CBCL internalizing scale. Own report: in 2017, an index of low self-esteem, not feeling happy, and feeling lonely in class; in 2024, the PHQ-4. Flat profile for girls: $p = 0.03$ for the 2024 PHQ-4 and above 0.2 for the other four measures. Different informants' reports of internalizing symptoms correlate at 0.25 on average (De Los Reyes et al., 2015).

▸ Table ▸ Profiles

Socioeconomic and family environment does not explain the gap

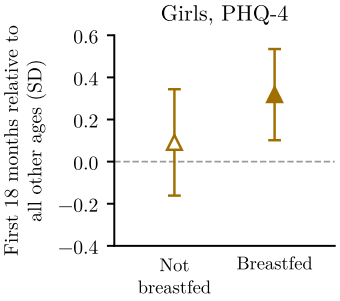
The earthquake worsened the environment (Gillmore, 2026), but not more for those exposed in the first 18 months

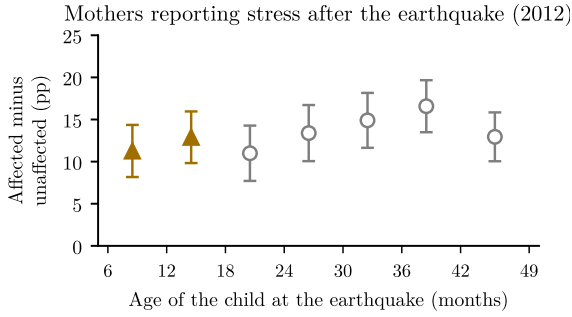


Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. In blue, the mechanisms in Gillmore (2026). Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack (*mediagua*) or a lent or shared dwelling. Girls exposed at 6–17 months relative to all other ages; 95% CI. The two estimates that differ from zero point to a better environment, not a worse one.

▸ Table

The effect is larger among breastfed girls, but imprecise





Breastfed, still breastfed on 27-F (2012 report): 53% of girls exposed at 6–17 months. Left, with six-month age group × breastfed fixed effects; the difference between breastfed and not breastfed is 0.23 SD (SE 0.17; wild bootstrap $p = 0.20$). Right, the share of mothers who reported stress, distress, or anxiety after the earthquake, unweighted; 95% CI. It did not rise more for the mothers of the youngest: for girls exposed at 6–17 months relative to all other ages, −3.3 points (SE 1.8).

▸ Table ▸ By age ▸ Maternal stress ▸ Means

Findings

- Among girls, the effect of the earthquake on mental health at ages 14–18 varies with age at exposure (flat profile rejected, $p = 0.03$); among boys, it does not ($p = 0.90$).
- Girls exposed in the first 18 months report more symptoms than those exposed later, by 0.21 SD, a comparison chosen after inspecting the profile.
- The gap is in the girls' own reports, significant in 2024 and of the same sign in 2017; their caregivers report no difference in 2012, 2017, or 2024.
- The socioeconomic and family environment did not worsen more for the youngest, and the stress their mothers reported in 2012 did not rise more.

Open questions

- Why only girls, and only in their own reports?

  Hypothesis: early stress leaves a latent vulnerability that surfaces in adolescence, when the gender gap in depression opens (McCrory and Viding, 2015; Salk, Hyde, and Abramson, 2017), as internalizing symptoms that caregivers seldom see.

- What happens with exposure before 6 months?

  Observed only in 2017, 209 girls: those exposed at 0–5 months score 0.30 SD higher on the caregiver report (SE 0.15) and 0.19 SD on their own (SE 0.12), relative to those exposed at ages 1–4.

- What does breastfeeding transmit, if the stress mothers reported did not rise more?

  Hypothesis: the mother's acute stress response in the months after the earthquake, which a report two years later captures only in part.

References

Almond, Douglas, Janet Currie, and Valentina Duque. 2018. “Childhood Circumstances and Adult Outcomes: Act II.” ***Journal of Economic Literature*** 56 (4): 1360–1446.

Andrabi, Tahir, Benjamin Daniels, and Jishnu Das. 2023. “Human Capital Accumulation and Disasters: Evidence from the Pakistan Earthquake of 2005.” ***Journal of Human Resources*** 58 (4): 1057–1096.

Conger, Rand D., and M. Brent Donnellan. 2007. “An Interactionist Perspective on the Socioeconomic Context of Human Development.” ***Annual Review of Psychology*** 58: 175–199.

De Los Reyes, Andres, Tara M. Augenstein, Mo Wang, Sarah A. Thomas, Deborah A. G. Drabick, Darcy E. Burgers, and Jill Rabinowitz. 2015. “The Validity of the Multi-Informant Approach to Assessing Child and Adolescent Mental Health.” ***Psychological Bulletin*** 141 (4): 858–900.

Gee, Dylan G., and B. J. Casey. 2015. “The Impact of Developmental Timing for Stress and Recovery.” ***Neurobiology of Stress*** 1: 184–194.

Gillmore, Roberto. 2026. “Natural Disasters and Early Child Development: Evidence from an Earthquake.” ***Economics of Education Review*** 115: 102817.

Kroenke, Kurt, Robert L. Spitzer, Janet B. W. Williams, and Bernd Löwe. 2009. “An Ultra-Brief Screening Scale for Anxiety and Depression: The PHQ-4.” ***Psychosomatics*** 50 (6): 613–621.

References

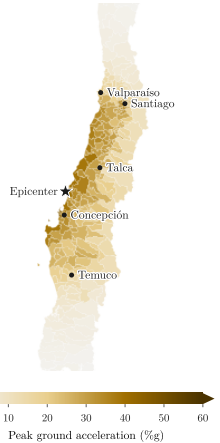
McCrory, Eamon J., and Essi Viding. 2015. “The Theory of Latent Vulnerability: Reconceptualizing the Link between Childhood Maltreatment and Psychiatric Disorder.” ***Development and Psychopathology*** 27 (2): 493–505.

Persson, Petra, and Maya Rossin-Slater. 2018. “Family Ruptures, Stress, and the Mental Health of the Next Generation.” ***American Economic Review*** 108 (4–5): 1214–1252.

Salk, Rachel H., Janet S. Hyde, and Lyn Y. Abramson. 2017. “Gender Differences in Depression in Representative National Samples: Meta-Analyses of Diagnoses and Symptoms.” ***Psychological Bulletin*** 143 (8): 783–822.

Torche, Florencia. 2011. “The Effect of Maternal Stress on Birth Outcomes: Exploiting a Natural Experiment.” ***Demography*** 48 (4): 1473–1491.

Figure A1—Peak Ground Acceleration by Municipality



Notes: Peak ground acceleration of the 27-F earthquake, in percent of gravity, at the centroid of each municipality. Affected municipalities in the main specification are those of the six regions with destructive shaking; the map shows how shaking varied within and across them. *Source:* USGS ShakeMap of the event.

Back

Table A1—Descriptive Statistics

	All		Mean by sex	
	(1) Mean	(2) SD	(3) Girls	(4) Boys
Panel A. Adolescent, 2024				
PHQ-4 score (0–12)	4.32	2.68	4.89	3.78
PHQ-4 of 6 or more (share)	0.27	0.44	0.34	0.19
Age in August 2024 (years)	16.76	1.05	16.76	16.76
Age at the earthquake (months)	27.06	12.63	27.07	27.06
Affected municipality (share)	0.78	0.42	0.78	0.78
Panel B. Birth and household, 2010				
Birth weight (kg)	3.39	0.50	3.34	3.44
Gestation (weeks)	38.64	2.01	38.69	38.59
Mother's age (years)	29.52	7.11	29.60	29.44
Mother's education (years)	10.31	3.18	10.28	10.34
Household size	4.93	1.68	4.92	4.93
Rural (share)	0.09	0.29	0.09	0.09
Panel C. After the earthquake, 2012				
Breastfed on 27-F (share)	0.24	0.43	0.24	0.24
Mother reported stress (share)	0.14	0.34	0.14	0.13
Adolescents	9,996		4,961	5,035

Notes: Children of the 2010 baseline (15,175) interviewed in 2024 (10,003) with a PHQ-4 score (9,996); ELPI sampling weights. Panel C was reported in 2012; breastfed means still breastfed on 27-F.

◂ Back

Table A2—Balance on Characteristics Fixed Before the Earthquake

	At birth (SD)			Mother, 2010 (years)	
	(1) Weight	(2) Length	(3) Gestation	(4) Education	(5) Age
6–17 months × Affected	0.014	−0.033	0.027	−0.275	−0.412
	(0.085)	(0.071)	(0.076)	(0.234)	(0.576)
Flat profile, joint test p-value	0.059	0.305	0.340	0.640	0.567
Observations	4,530	4,525	4,818	4,882	4,920
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No

Notes: Girls. Each column regresses a characteristic fixed before the earthquake on ages 6–17 months × affected, with municipality, birth-year, and six-month age-group fixed effects and no controls; the joint test is of a flat profile across the seven six-month age groups. Birth weight, length, and gestation are standardized; the mother's education and age were reported in 2010. Among the 4,190 girls with birth records, the PHQ-4 coefficient for those exposed at 6–17 months, relative to all other ages, is 0.188 (0.079) without and 0.188 (0.079) with the three birth outcomes as controls. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

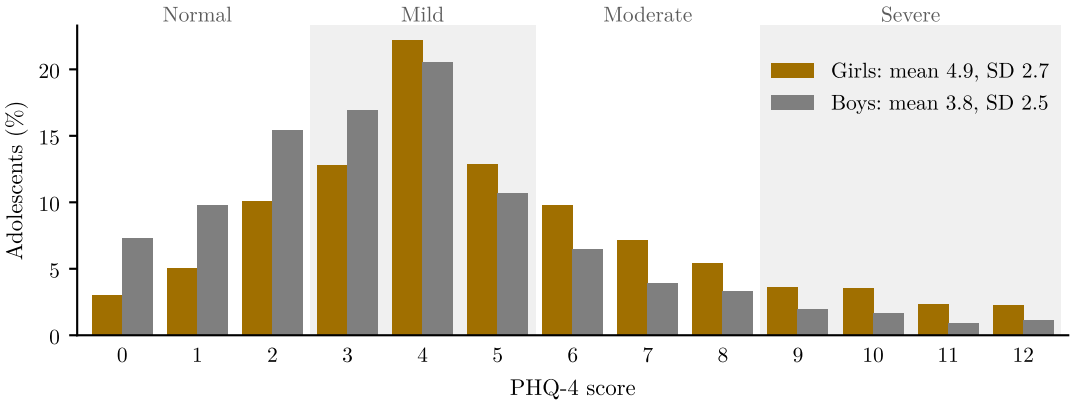
Table A3—Attrition between 2010 and 2024

	Interviewed in 2024			Birth weight, girls (SD)	
	(1)	(2)	(3)	(4)	(5)
	Girls	Boys	All	2010 baseline	Interviewed in 2024
6–17 months × Affected	−0.039	−0.012	−0.027	0.019	0.004
	(0.029)	(0.029)	(0.020)	(0.065)	(0.079)
Flat profile, joint test p-value	0.757	0.562	0.522	0.649	0.500
Mean of dependent variable	0.67	0.67	0.67		
Observations	6,990	7,171	14,161	6,409	4,300
Region FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes	Yes	Yes

Notes: Children with the 2010 developmental assessment, which gives age in months (14,161 of 15,175). Columns 1–3 equal one if interviewed in 2024; columns 4–5, standardized birth weight of all girls assessed in 2010 and of those interviewed in 2024. Age at the earthquake is the age at the 2010 assessment minus the months from 27-F to the region's median assessment month, for stayers and leavers alike. Joint test as in Table A2. The 2010 files do not identify the municipality, so region fixed effects, no weights, and robust standard errors.

◂ Back

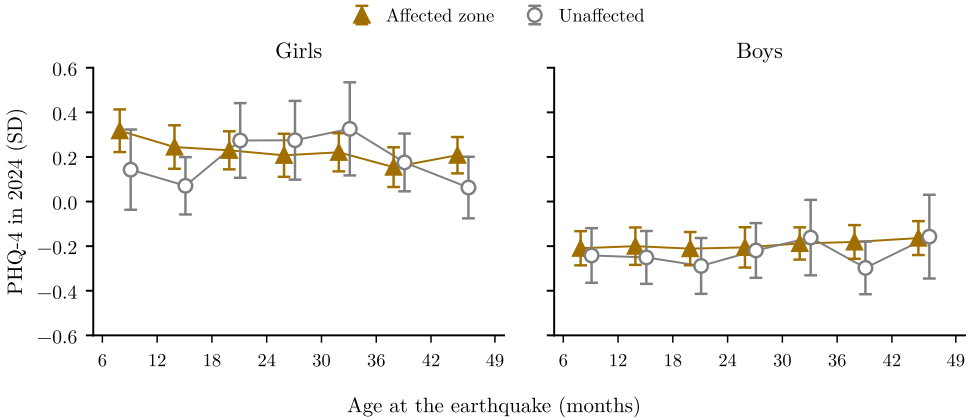
Figure A2—Distribution of the PHQ-4 Score



Notes: PHQ-4 score in 2024, weighted with ELPI sampling weights. The score adds four items scored 0 to 3; Kroenke et al. (2009) classify 0–2 as normal, 3–5 as mild, 6–8 as moderate, and 9–12 as severe.

Back

Figure A3—PHQ-4 by Age at the Earthquake and Zone



Notes: Weighted mean of the standardized PHQ-4 in 2024 by six-month age group and zone; 95% CI clustered by municipality of selection. Among girls, the affected zone is nearly flat except at 6–11 months; much of the profile comes from the unaffected zone, where girls exposed at 6–17 months score low and those exposed at 18–35 months score high. Relative to ages 36–49 months only, the gap at 6–17 months is 0.10 SD (SE 0.09).

Back

Table A4—Effect on Mental Health by Age at the Earthquake

	PHQ-4 (SD)		
	(1) All	(2) Girls	(3) Boys
6–11 months	0.069	0.161	−0.014
	(0.049)	(0.083)	(0.071)
12–17 months	0.075	0.143	0.015
	(0.057)	(0.079)	(0.073)
18–23 months	−0.033	−0.090	0.032
	(0.063)	(0.088)	(0.084)
24–29 months	−0.055	−0.131	−0.014
	(0.057)	(0.073)	(0.071)
30–35 months	−0.103	−0.151	−0.049
	(0.058)	(0.086)	(0.086)
36–41 months	0.011	−0.044	0.070
	(0.049)	(0.073)	(0.065)
42–49 months	0.036	0.113	−0.040
	(0.054)	(0.074)	(0.092)
Flat profile, joint test p-value	0.290	0.029	0.905
Observations	9,996	4,961	5,035
Municipality FE	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Notes: Affected minus unaffected municipalities in the standardized PHQ-4 in each six-month age group, normalized to average zero; the joint test is of equal effects at all ages. Controls and weights as in Table A5. For girls, the joint test gives $p = 0.03$ with month-of-birth fixed effects and $p = 0.04$ with peak ground acceleration instead of the zone. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

Back

Table A5—The First 18 Months Relative to All Other Ages

	PHQ-4 (SD)				
	(1) All	(2) Girls	(3) Boys	(4) Girls	(5) Girls
6–17 months × Affected	0.097	0.206	0.001	0.214	0.072
	(0.049)	(0.075)	(0.064)	(0.076)	(0.025)
Wild bootstrap p-value	0.058	0.010	0.991	0.008	0.006
Observations	9,996	4,961	5,035	4,961	4,961
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	No	Yes
Six-month age-group FE	Yes	Yes	Yes	No	Yes
Month-of-birth FE	No	No	No	Yes	No
Exposure	Zone	Zone	Zone	Zone	PGA
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Affected minus unaffected municipalities in the standardized PHQ-4 at ages 6–17 months relative to all other ages, a comparison chosen after inspecting Table A4. Controls are a linear trend in month of birth, sex, the mother's age and education, household size, rural residence (2010), the mother's number of children, and family history of mental illness (2012). Column 4 uses month-of-birth fixed effects; column 5, peak ground acceleration per 10 points of percent of g (28 more in the average affected municipality). Wild bootstrap with 9,999 Rademacher draws under the null. For girls, randomization inference over the 5,005 ways of choosing six of the 15 regions as affected gives $p = 0.06$; with standard errors clustered by region, $p = 0.02$. Girls minus boys, 0.205 (0.099), from a model with every term interacted with sex. ELPI weights. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A6—Other Outcomes, Girls

	Adolescent report (SD)			Caregiver report, CBCL (SD)	
	(1) PHQ-4	(2) Substance use	(3) Victimization	(4) Internalizing	(5) Externalizing
6–17 months × Affected	0.206	0.077	0.006	0.085	0.077
	(0.075)	(0.074)	(0.076)	(0.082)	(0.073)
Flat profile, joint test p-value	0.029	0.548	0.007	0.638	0.337
Observations	4,961	4,961	4,539	4,748	4,748
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Girls. Each column is a separate regression with the specification of Table A5. The adolescent reports the PHQ-4, substance use (any alcohol, tobacco, cannabis, or other drug in the last 12 months), and peer victimization; the caregiver reports internalizing and externalizing problems (CBCL). All measured in 2024 and standardized; higher is worse. Boys in Table A7. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A7—Other Outcomes, Boys

	Adolescent report (SD)			Caregiver report, CBCL (SD)	
	(1) PHQ-4	(2) Substance use	(3) Victimization	(4) Internalizing	(5) Externalizing
6–17 months × Affected	0.001	0.110	0.018	0.034	0.086
	(0.064)	(0.072)	(0.077)	(0.070)	(0.060)
Flat profile, joint test p-value	0.905	0.181	0.097	0.871	0.546
Observations	5,035	5,034	4,637	4,804	4,804
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Boys. Outcomes and specification as in Table A6. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

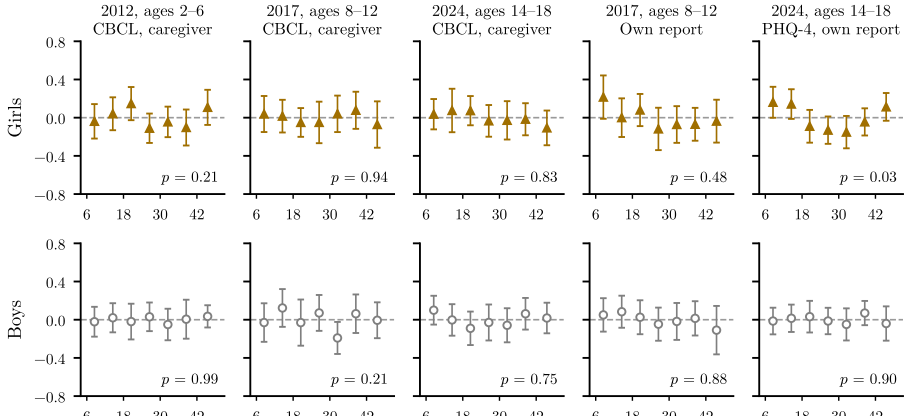
Table A8—Mental Health in Three Rounds

	Caregiver, CBCL internalizing (SD)			Own report (SD)	
	(1) 2012 ages 2–6	(2) 2017 ages 8–12	(3) 2024 ages 14–18	(4) 2017 ages 8–12	(5) 2024 ages 14–18
Panel A. Girls					
6–17 months × Affected	0.000	0.039	0.081	0.150	0.206
	(0.089)	(0.097)	(0.078)	(0.123)	(0.075)
Flat profile, joint test p-value	0.215	0.944	0.833	0.479	0.029
Observations	4,053	3,533	4,946	3,510	4,961
Panel B. Boys					
6–17 months × Affected	−0.003	0.059	0.064	0.098	0.001
	(0.073)	(0.116)	(0.069)	(0.068)	(0.064)
Flat profile, joint test p-value	0.987	0.211	0.748	0.881	0.905
Observations	4,087	3,650	5,024	3,603	5,035
Municipality FE	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: The adolescents interviewed in 2024, in each round in which they have the measure; specification of Table A5, with 2024 weights. Caregiver: CBCL internalizing scale, standardized. Own report: in 2017, the mean of three standardized items (low self-esteem, not feeling happy, feeling lonely in class); in 2024, the PHQ-4. Higher is worse. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Figure A4—The Age Profile in Three Rounds



Notes: For each measure of Table A8, affected minus unaffected in each six-month age group, normalized to average zero; 95% CI. *p*, joint test of a flat profile.

Back

Table A9—The Socioeconomic and Family Environment

	Mechanisms in Gillmore (2026), SD			Other household conditions, SD		
	(1) Lower income	(2) Mother not employed	(3) Damaged housing	(4) Conflict at home	(5) Overcrowded housing	(6) Precarious housing
6–17 months × Affected	−0.013	−0.188	−0.005	−0.152	0.000	0.051
	(0.064)	(0.065)	(0.049)	(0.073)	(0.065)	(0.061)
Flat profile, joint test p-value	0.880	0.088	0.944	0.027	0.366	0.615
Observations	4,478	4,925	4,423	4,701	4,961	4,960
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Girls. Each column is a standardized index of the household environment; higher is worse. Measured in 2010, one to nine months after the earthquake, except housing damage, reported in 2012. Lower income, per capita; damaged housing, destroyed or with major damage; conflict at home, the mother shouting, being hostile, or hitting during the visit; overcrowded housing, persons per bedroom; precarious housing, an emergency shack or a lent or shared dwelling. Specification of Table A5. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

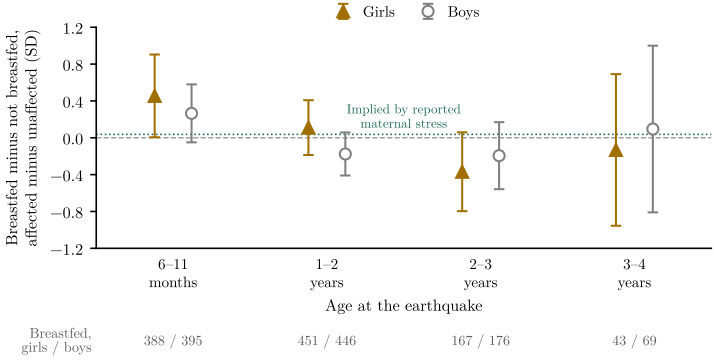
Table A10—Breastfeeding and the Effect in the First 18 Months

	PHQ-4 (SD)		
	(1) Girls	(2) Girls	(3) Boys
6–17 months × Affected × breastfed	0.318	0.295	0.082
	(0.110)	(0.110)	(0.091)
6–17 months × Affected × not breastfed	0.091	0.081	−0.045
	(0.129)	(0.131)	(0.081)
Difference, breastfed − not breastfed	0.227	0.214	0.127
	(0.167)	(0.167)	(0.112)
Wild bootstrap p-value, difference	0.200	0.225	0.285
Breastfed, share at 6–17 months	0.53	0.54	0.51
Observations	4,409	4,338	4,465
Municipality FE	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes
Six-month age group × breastfed FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Mother's education and employment × breastfed and × affected	No	Yes	No

Notes: Difference between affected and unaffected municipalities in the standardized PHQ-4 for those aged 6–17 months, relative to all other ages, separately for those still breastfed on 27-F (2012 report) and those who were not. Column 2 lets breastfeeding and the affected zone vary with the mother's education and employment (2010). Controls, weights, and wild bootstrap as in Table A5. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Figure A5—Breastfeeding and the Effect of the Earthquake by Age



Notes: For each age at the earthquake, the difference between affected and unaffected municipalities among those breastfed on 27-F minus the same difference among those who were not, with fixed effects for age group × breastfed, municipality, and birth year, and the controls of Table A5. The dotted line is the value implied by maternal stress reported in 2012 (0.29 × 0.13). 95% CI.

Back

Table A11—Maternal Stress, Breastfeeding, and Daughters' Mental Health

	Daughters' PHQ-4 (SD)		
	(1) Baseline	(2) Mother's SES	(3) Mother's SES and 6–17 months
Maternal stress × not breastfed	−0.026	−0.006	−0.024
	(0.056)	(0.064)	(0.069)
Maternal stress × breastfed	0.294	0.284	0.213
	(0.088)	(0.096)	(0.119)
Difference, breastfed − not breastfed	0.320	0.290	0.237
	(0.108)	(0.101)	(0.109)
Observations	4,409	4,377	4,377
Municipality FE	Yes	Yes	Yes
Birth-year FE	Yes	Yes	Yes
Six-month age-group FE	Yes	Yes	Yes
Six-month age group × breastfed FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Stress and breastfed × mother's education and employment	No	Yes	Yes
Stress × 6–17 months	No	No	Yes

Notes: Girls, all ages. Maternal stress equals one if the mother reported stress, distress, or anxiety after the earthquake (2012); breastfed equals one if the child was still breastfed on 27-F (2012 report). Column 2 lets stress and breastfeeding vary with the mother's education and employment (2010); column 3 also lets stress vary at 6–17 months. Controls and weights as in Table A5. Standard errors clustered by municipality of selection (116 clusters) in parentheses.

◂ Back

Table A12—Maternal Stress and Breastfeeding by Age at the Earthquake

	Mothers with stress (%)		Breastfed (%)
	(1) Affected	(2) Unaffected	(3) On 27-F
6–11 months	15.0	3.8	61.8
	(1.2)	(1.1)	(1.4)
12–17 months	16.1	3.2	42.6
	(1.2)	(1.0)	(1.4)
18–23 months	15.0	4.0	30.3
	(1.2)	(1.2)	(1.3)
24–29 months	17.5	4.1	15.6
	(1.2)	(1.2)	(1.0)
30–35 months	18.1	3.2	12.9
	(1.3)	(1.0)	(1.0)
36–41 months	19.2	2.6	5.0
	(1.3)	(0.9)	(0.6)
42–49 months	16.6	3.7	3.3
	(1.1)	(1.0)	(0.5)
Observations	6,772	2,132	8,874

Notes: Unweighted means, with standard errors in parentheses, by the child's age at the earthquake, among the adolescents of Table A1 whose mother answered each question in 2012. Columns 1 and 2 report the share of mothers who reported stress, distress, or anxiety after the earthquake, in affected and unaffected municipalities; column 3, the share of children still breastfed on 27-F.

◂ Back